Riddling

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Chapter 1

Algebra

1.1 50 dollars spent

I have 50 dollars and I spend them in four purchases. After each purchase I write down how much I spent and how much I have left:

I spend	Left
20	30
15	15
9	6
6	0
50	51

The first column sums to 50, as it should, but the second column sums to 51. Where did the extra dollar come from?

1.1.1 Solution

There is no extra dollar. The sum of the second column has no reason to be equal to 50 and the fact that it is close to 50 is a coincidence.

The second column is a list of balances, and adding up balances at different points in time does not measure anything. If I spent the whole 50 dollars at once the second column would sum to 0. If I spent the 50 dollars one dollar at a time the second column would sum to $49 + 48 + \cdots + 1 + 0 = \frac{49 \cdot 50}{2} = 1225$. Only the first column has to sum to 50.

1.2 Area of a circle

Prove the formula for the area of a circle by integrating the equation of a circle

1.2.1 Solution

The equation of a circle with radius r centered at the origin is $x^2 + y^2 = r^2$. In the first quadrant (where x and y are both positive), we can solve for y:

$$y^2 = r^2 - x^2$$
$$y = \sqrt{r^2 - x^2}$$

The integral represents the area of one-quarter of the circle.

$$q = \int_0^r \sqrt{r^2 - x^2}\, dx$$

To get the area of the full circle, we multiply the area of the quarter circle by 4.

$$a = 4 \int_0^r \sqrt{r^2 - x^2}\, dx$$

To solve this integral, we use the trigonometric substitution:

$$x = r \sin(\theta)$$
$$dx = r \cos(\theta)\, d\theta$$

Change of Limits: When $x = 0$, $\theta = 0$ When $x = r$, $\theta = \frac{\pi}{2}$
Substituting into the Integral:

$$\begin{aligned} a &= 4 \int_0^{\frac{\pi}{2}} \sqrt{r^2 - (r \sin(\theta))^2} \cdot r \cos(\theta)\, d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} \sqrt{r^2 - r^2 \sin^2(\theta)} \cdot r \cos(\theta)\, d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} \sqrt{r^2 (1 - \sin^2(\theta))} \cdot r \cos(\theta)\, d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} \sqrt{r^2 \cos^2(\theta)} \cdot r \cos(\theta)\, d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} r \cos(\theta) \cdot r \cos(\theta)\, d\theta \\ &= 4r^2 \int_0^{\frac{\pi}{2}} \cos^2(\theta)\, d\theta \end{aligned}$$

We use the trigonometric identity:

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

$$\begin{aligned} a &= 4r^2 \int_0^{\frac{\pi}{2}} \frac{1 + \cos(2\theta)}{2}\, d\theta \\ &= 2r^2 \int_0^{\frac{\pi}{2}} (1 + \cos(2\theta))\, d\theta \end{aligned}$$

Integrating

$$\begin{aligned} a &= 2r^2 \left[\theta + \frac{\sin(2\theta)}{2}\right]_0^{\frac{\pi}{2}} \\ &= 2r^2 \left[\left(\frac{\pi}{2} + 0\right) - (0 + 0)\right] \\ &= \pi r^2 \end{aligned}$$

We have successfully proven that the area of a circle with radius r is πr^2.

1.3 Clock hands meeting

How many times a day do the hour hand and the minute hand of a clock meet?

1.3.1 Solution

If you thought it was 24, you are mistaken: they meet 22 times.

The hands meet at 12:00, then a little after 1:05, then a little after 2:10 and so on. But between 11:00 and 12:00 they do not meet, since the meeting that would be "a little after 11" is the meeting at 12:00. So in 12 hours they meet 11 times (counting the 12:00 at the start but not the 12:00 at the end), and in a day they meet 22 times.

Another way to count: in a day the minute hand goes around the clock 24 times and the hour hand twice. Every meeting is a time when the minute hand overtakes the hour hand by a full lap, and the minute hand gains $24 - 2 = 22$ laps in a day. So they meet 22 times, every $\frac{24}{22} = 1\frac{1}{11}$ hours.

1.4 The drying watermelon

A watermelon weighs 10 kg and is 99% water. It sits in the sun and some of the water evaporates, until it is only 98% water. How much does the watermelon weigh now?

1.4.1 Solution

It weighs 5 kg, half of what it weighed before.

The watermelon is 99% water, so the part that is not water is 1% of 10 kg, which is 0.1 kg. Only water evaporates, so the part that is not water still weighs 0.1 kg, but now it is 2% of the watermelon. 2% is $\frac{1}{50}$, so the watermelon weighs $50 \cdot 0.1 = 5$ kg.

The answer looks surprising because a change from 99% to 98% sounds small. But the part that is not water went from 1% to 2% of the weight, which means that the weight was cut in half.

1.5 Equation involving roots

Solve the equation $\sqrt{x} + \sqrt{x+87} = 87$.

1.5.1 Solution

Let $a = \sqrt{x}$ and $b = \sqrt{x+87}$. Then $a + b = 87$ and $b^2 - a^2 = 87$. Since $b^2 - a^2 = (b-a)(b+a)$ we get $(b-a) \cdot 87 = 87$, so $b - a = 1$. Together with $a + b = 87$ this gives $a = 43$ and $b = 44$, so $x = 43^2 = 1849$.

Indeed $\sqrt{1849} + \sqrt{1849+87} = \sqrt{1849} + \sqrt{1936} = 43 + 44 = 87$.

The same thing can be seen from the list of squares $1, 4, 9, 16, \ldots$ whose differences are the odd numbers $3, 5, 7, \ldots$: the difference between n^2 and $(n+1)^2$ is $2n + 1$, and $87 = 2 \cdot 43 + 1$ is the difference between 43^2 and 44^2.

1.5.2 Follow up

Solve $\sqrt{x} + \sqrt{x+87} = \sqrt{87}$.

1.5.3 Solution

$x = 0$ is a solution. There are no others: for every $x > 0$ we have $\sqrt{x} > 0$ and $\sqrt{x+87} > \sqrt{87}$, so the left side is larger than $\sqrt{87}$.

1.6 Infinite nested root

What is the value of $\sqrt{2+\sqrt{2+\sqrt{2+\ldots}}}$?

1.6.1 Solution

Call the whole expression x. The expression under the first root is 2 plus the whole expression again, so

$$\begin{gathered}\sqrt{2+x} = x \\ 2 + x = x^2 \\ x^2 - x - 2 = (x-2)(x+1) = 0\end{gathered}$$

The solutions are $x = 2$ and $x = -1$. A square root is never negative, so $x = 2$.

To be careful, we should also check that the expression has a value at all. It is the limit of the sequence $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{2+a_n}$. If $a_n < 2$ then $a_{n+1} = \sqrt{2+a_n} < \sqrt{4} = 2$, and also $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2-a_n)(1+a_n) > 0$. So the sequence is increasing and bounded by 2, and therefore it has a limit, which by the calculation above is 2.

1.7 Into the night

This riddle was invented after watching the Netflix series "Into the Night".

Assume that the circumference of the earth along the equator is 48,000 km, that a plane flies at 1,000 km/h, and that night is exactly half of the day. You fly along the equator and want to stay in the dark for as long as possible. Should you fly east or west, and for how long and how far can you fly?

1.7.1 Solution

Night covers half of the equator, a band of 24,000 km. The earth turns once a day, so this band moves west along the equator at $48000/24 = 2000$ km/h. Its western edge is dusk and its eastern edge is dawn. In both directions it is best to take off at dusk, when the whole band of night is still ahead of the sun.

Flying east. Dawn is 24,000 km east of you and moves toward you at 2,000 km/h while you fly toward it at 1,000 km/h. If you fly for n hours then $1000n + 2000n = 24000$, so $n = 8$ and you cover 8,000 km.

Flying west. Dusk runs away from you, since it moves west faster than you do. Dawn is 24,000 km behind you and catches up at $2000 - 1000 = 1000$ km/h. So $2000n = 1000n + 24000$, which gives $n = 24$ and you cover 24,000 km.

So you should fly west: you can stay in the dark for a whole day and cover half of the globe. Going around the globe in the dark takes two such flights when flying west, but six when flying east.

1.8 The number one half

A teacher writes 96 fractions on the board: $\frac{1}{96}, \frac{2}{96}, \frac{3}{96}, \ldots, \frac{95}{96}, \frac{96}{96}$. A student then plays a game: in every turn he picks any two numbers a and b on the board, erases them, and writes $2ab - a - b + 1$ instead. After 95 turns only one number is left. What is it?

1.8.1 Solution

The last number is $\frac{1}{2}$, no matter how the student plays.

Let $f(a, b) = 2ab - a - b + 1$. Then $f(\frac{1}{2}, b) = b - \frac{1}{2} - b + 1 = \frac{1}{2}$, and in the same way $f(a, \frac{1}{2}) = \frac{1}{2}$. The number $\frac{48}{96} = \frac{1}{2}$ is on the board from the start. Whenever it is picked, the new number is $\frac{1}{2}$ again, so there is always a $\frac{1}{2}$ on the board, and therefore the last number is $\frac{1}{2}$.

A cleaner way to see what is going on: $2f(a, b) - 1 = 4ab - 2a - 2b + 1 = (2a - 1)(2b - 1)$. So if we look at the numbers $2x - 1$ for all the numbers x on the board, every turn replaces two of them by their product, and the product of all of them never changes. At the start one of them is $2 \cdot \frac{1}{2} - 1 = 0$, so the product is 0, and at the end the last number x satisfies $2x - 1 = 0$.

This is also why the riddle uses an even number like 96: with an odd number of fractions, $\frac{1}{2}$ would not be on the board.

1.9 Square root of i

What is the square root of i?

1.9.1 Solution

Let's assume $\sqrt{i} = a + b * i$

1.10 Working together

Alice and Bob together do a certain job in 2 hours. Alice and Charlie together do it in 3 hours. Bob and Charlie together do it in 4 hours. How long would it take Alice, Bob and Charlie to do the job together?

1.10.1 Solution

Let A, B and C be the parts of the job that Alice, Bob and Charlie do in an hour. Then

$$A + B = \tfrac{1}{2}, \quad A + C = \tfrac{1}{3}, \quad B + C = \tfrac{1}{4}$$

There is no need to find A, B and C separately. Adding the three equations, every person appears twice:

$$2(A + B + C) = \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} = \tfrac{13}{12}$$

so $A + B + C = \frac{13}{24}$. Working together they do $\frac{13}{24}$ of the job every hour, so the job takes them $\frac{24}{13}$ hours, which is about 1 hour and 51 minutes.

Chapter 2

Algorithms

2.1 25 horses, 3 best

You need to find the 3 fastest horses out of 25. You can only race 5 at a time and you don't have a timer. What's the minimum number of races you need and who will you race?

2.1.1 Solution

Let's mark all horses as H_1 - H_{25}. First race 5 races H_1-H_5, H_6-H_{10}, H_{11}-H_{15}, H_{16}-H_{20}, H_{21}-H_{25}. Denote the winners as $W_{1,5}$, $W_{6,10}$, $W_{11,15}$, $W_{16,20}$, $W_{21,25}$. Second runners up will be $SRU_{1,5}$, $SRU_{6,10}$, $SRU_{11,15}$, $SRU_{16,20}$, $SRU_{21,25}$. Third runners up will be $TRU_{1,5}$, $TRU_{6,10}$, $TRU_{11,15}$, $TRU_{16,20}$, $TRU_{21,25}$. We will race all the winners and get WW_1 - WW_5.

Now let's use some logic.

WW_1 is the fastest around so we don't need to race him again and he will be the top in the top 3.

Discard anyone in the group that got won by WW_4 and WW_5 since they got beat by at least three horses (WW_1, WW_2, WW_3) so they, and by inference anyone they beat, cannot be in the top 3.

Discard anyone in the group that got won by WW_3 but not WW_3 himself. This is because there are at least two horses better than WW_3 so anyone that WW_3 beat cannot be in the top 3 but WW_3 maybe the third in the entire group. So WW_3 joins the final race.

In the group of horses won by WW_2 we can discard everyone but WW_2 and the second runner up. Third runner up and down have at least 3 horses better than them (SRU_{WW_2}, WW_2 and WW_1). So WW_2 and SRU_{WW_2} joins the race.

In the group of WW_1 we can discard everyone but the second and third runners up. So SRU_{WW_1} and TRU_{WW_1} join the race.

So now we will race WW_3, WW_2, SRU_{WW_2}, SRU_{WW_1}, TRU_{WW_1}. The first two will join WW_1 in being the fastest 3.

Number of races: 7

Why can't it be done in 6? Imagine that the results are decided by a spiteful demon, who only has to stay consistent with some real order of the horses. Call a horse *unbeaten* as long as it has not finished behind any other horse. An unbeaten horse could still be the fastest, so when we are done exactly one horse may be unbeaten. The demon's rule is that in every race the unbeaten horses finish ahead of the rest. This never contradicts anything already known, since no horse is known to be faster than an unbeaten one. So a race with k unbeaten horses beats at most $k - 1 \leq 4$ of them for the first time, and we need 24 first defeats, which takes at least 6 races. And with only 6 races, every race must be made of 5 unbeaten horses, so a horse that has been beaten never races again.

Now look at the 6th race. The first 5 races beat 20 different horses, and the winner of the 5th race is not one of them, so at least 21 horses raced before the 6th race and at most 4 did not. So some horse in the 6th race has raced before, and since it is unbeaten it won every race it ran. The demon lets it win the 6th race too; call it C. Let Q be the runner-up of an earlier race won by C, and R and T the second and third horses of the 6th race. Q finished behind C only and never raced again, and R and T never lost before the 6th race, so nothing links Q to R or T. Both orders C, Q, R, T and C, R, T, Q (with all the other horses behind) agree with every result, and they have different top 3s. So 6 races are never enough, and 7 is the minimum.

2.2 Building a crack free wall

This riddle is from Project Euler (problem 215).

You need to build a wall 32 units wide and 10 units high out of bricks of size 2×1 and 3×1, laid horizontally. To make the wall strong, no two bricks in adjacent rows may end at the same place, except at the two ends of the wall: the gaps between bricks must never line up to form a running crack. In how many ways can the wall be built?

2.2.1 Solution

The rule cannot apply to the whole wall at once: a row of width 32 has at least 10 gaps between bricks, and there are only 31 places for them, so ten rows cannot all have different gaps. That is why the rule is only about adjacent rows.

First list all the possible rows. A row is a sequence of 2s and 3s that adds up to 32, and it is described by the set of places where its gaps are. The number of rows a_n of width n satisfies $a_n = a_{n-2} + a_{n-3}$, since the last brick has length 2 or 3, and for width 32 there are 3329 rows. Represent the set of gaps of each row as a bit mask. Two rows can lie on top of each other exactly when the bitwise and of their masks is 0.

Now count the walls row by row. Let $c_h(r)$ be the number of walls of height h whose top row is r. Then $c_1(r) = 1$ for every row and

$$c_{h+1}(r) = \sum_{s \text{ compatible with } r} c_h(s)$$

The answer is the sum of $c_{10}(r)$ over all rows r. Each step goes over all the pairs of compatible rows, which is at most 3329^2, about eleven million, so this is fast. The number of walls is 806,844,323,190,414.

As a check, the same method gives 8 walls of width 9 and height 3. Note that this method depends on the number of possible rows being small: a wall 100 units wide would have about $6.7 \cdot 10^{11}$ possible rows.

2.3 Estimating π

Write a program that gives better and better estimates of π without using a series like the Taylor series. The program should be easy to write and deterministic: the longer it runs, the better the estimate is guaranteed to be.

2.3.1 Solution

A popular answer is to pick random points in a square of side 1 and count how many of them fall inside the circle of diameter 1 drawn in it. The circle has area $\pi/4$ and the square has area 1, so 4 times the fraction of points inside the circle estimates π. But this does not meet the requirements: it depends on a good random number generator, and more samples only make a good estimate likely, they do not guarantee it.

Instead, count squares deterministically. Cover the quarter of the circle of radius N that lies in the first quadrant with a grid of $N \times N$ unit squares. The quarter circle has area $\frac{\pi N^2}{4}$. Let L be the number of squares that lie completely inside the circle: the square whose far corner is (i, j) is inside when $i^2+j^2 \leq N^2$. Let U be the number of squares that touch the inside of the circle: the square whose near corner is (i, j) touches it when $i^2 + j^2 < N^2$. Then

$$\frac{4L}{N^2} \leq \pi \leq \frac{4U}{N^2}$$

The squares counted in U but not in L are the ones that the arc of the circle crosses. Going along the arc, every such square starts one step to the right or one step down from the previous one, so there are fewer than $2N$ of them and the two bounds are less than $\frac{8}{N}$ apart.

Counting the squares is easy: in column i there are $\lfloor\sqrt{N^2 - i^2}\rfloor$ squares completely inside the circle. Run the program with $N = 10, 100, 1000, \ldots$ and you get guaranteed bounds that improve with every step: $N = 1000$ gives $3.1375 \leq \pi \leq 3.1456$ and $N = 10000$ gives $3.1411 \leq \pi \leq 3.1420$.

2.4 Find number of bits in large data set

You get an array of 10000 32 bit integers and need to find out how many bits are turned on in the entire array. You are allowed to do precomputation and prepare any data structure of any size you see fit. What will you do? What will you do in practice?

2.4.1 Solution

The idea is to prepare a 2^{32} array of integers where for each place n the array contains the number of bits which are on for the number n. Then the computation is just a matter of iteration and summing up.

In practice this is a bad idea since the 2^{32} array is too large and could cause cache misses. Better is to hold an array of size 2^{16} or maybe even 2^8 so as to make a cache hit all the time for L1 cache. The size of the array should depend on the size of cache you want to keep hitting and the tradeoff between cache hit and doing more inside the loop should be determined by tests in practice.

2.5 Finding a good computer

There are 100 computers. At least 51 of them are good and the rest, at most 49, are broken. You may ask computer A whether computer B is good or broken. If A is good its answer is correct, but if A is broken its answer can be anything. Find a good computer while asking as few questions as possible in the worst case.

2.5.1 Solution

First set one computer aside and never use it. Among the remaining 99 computers at least 50 are good and at most 49 are broken, so the good computers are still a strict majority.

We build a chain of computers in which every computer has said that the next one is good. Go over the 99 computers one by one. For each computer x: if the chain is empty, put x on the chain without asking anything. Otherwise ask the computer at the top of the chain about x. If it says that x is good, put x at the top of the chain. If it says that x is broken, remove the top of the chain and throw away both it and x. When all the computers are processed, the computer at the top of the chain is good.

Why does this work?

1. If a good computer is on the chain then the computer after it on the chain is good too, since a good computer vouched for it. So the chain is made of some broken computers followed by some good computers, and if there is any good computer on the chain then the top is good.

2. Whenever we throw away two computers, at least one of them is broken: if the top of the chain is good then its answer is correct and x is broken, and if it is not good then it is broken itself. So throwing away a pair never hurts the majority: among the computers that were not thrown away the good ones remain a strict majority.

3. At the end all the computers that were not thrown away are on the chain. They have a good majority so the chain is not empty and holds a good computer, which means that its top is good.

Every computer except the first costs at most one question, so this protocol asks at most 98 questions.

2.6 Knight moves on a keypad

You are given a standard phone keypad:

```
1 2 3
4 5 6
7 8 9
  0
```

A chess knight stands on one of the keys. Write a function

`knight_combinations(from, n)`

that returns the number of different sequences of n knight moves that start at the key `from`. For example, from the key 1 there is 1 sequence of 0 moves, there are 2 sequences of 1 move: $(1, 6)$ and $(1, 8)$, and there are 5 sequences of 2 moves: $(1, 6, 1)$, $(1, 6, 7)$, $(1, 6, 0)$, $(1, 8, 1)$ and $(1, 8, 3)$.

2.6.1 Solution

First write down where a knight can go from each key:

```
0 -> 4, 6        5 -> nowhere
1 -> 6, 8        6 -> 1, 7, 0
2 -> 7, 9        7 -> 2, 6
3 -> 4, 8        8 -> 1, 3
4 -> 3, 9, 0     9 -> 2, 4
```

A sequence of n moves from key k is a first move to a neighbor j followed by a sequence of $n-1$ moves from j, so

$$f(k, 0) = 1, \quad f(k, n) = \textstyle\sum_{j \text{ a neighbor of } k} f(j, n-1)$$

Computing this naively with recursion takes exponential time, but computing $f(\cdot, 1), f(\cdot, 2), \ldots, f(\cdot, n)$ one after the other, each time for all 10 keys, takes $O(n)$ time and $O(1)$ memory.

We can do even better. Let M be the 10×10 adjacency matrix of the keys: $M_{kj} = 1$ if a knight can move from k to j and 0 otherwise. The entry $(M^n)_{kj}$ is the number of sequences of n moves from k to j, so the answer is the sum of row k of M^n. Raising a matrix to the power n by repeated squaring takes $O(\log n)$ matrix multiplications, so this solution takes $O(\log n)$ time. For example, there are 3728 sequences of 10 moves starting from the key 1.

2.7 Printing a linked list in reverse

Print a singly linked list of N elements from its end to its start, using only $O(1)$ memory. You may not change the list. Make the running time as small as you can.

2.7.1 Solution

The simplest solution is to walk from the head to the last element and print it, then walk from the head to the element before last and print it, and so on. This takes $O(N^2)$ time. Let's call it the naive method; it prints any segment of m elements backwards in $O(m^2)$ time.

First improvement. Split the list into $\sqrt{N}$ blocks of $\sqrt{N}$ elements each. Go over the blocks from the last to the first. For each block, walk from the head to the start of the block, which takes $O(N)$, and print the block backwards with the naive method, which takes $O((\sqrt{N})^2) = O(N)$. There are $\sqrt{N}$ blocks, so the total time is $O(\sqrt{N} \cdot N) = O(N^{1.5})$.

Second improvement. Split the list into $N^{1/3}$ blocks of $N^{2/3}$ elements each, and print each block backwards with the first improvement instead of the naive method. A block of $m = N^{2/3}$ elements takes $O(m^{1.5}) = O(N)$ time to print, and walking to the start of the block also takes $O(N)$. There are $N^{1/3}$ blocks, so the total time is $O(N^{4/3})$.

And so on. Repeating the trick k times gives a running time of $O(N^{1+1/k})$. Every level of the trick needs only a few variables, so for any fixed k the memory is still $O(1)$, and the running time is $O(N^{1+\epsilon})$ for any $\epsilon > 0$ we like.

If we allow $O(\log N)$ memory, we can go all the way: split the list into two halves, print the second half backwards and then the first half backwards, recursively. Every level of the recursion walks over the list once, and there are $\log N$ levels, so this takes $O(N \log N)$ time.

2.8 Random element of a list

This riddle is from Yaron Dorman.

You are given a pointer to a linked list whose length you do not know. Select one of its elements at random, such that every element has the same chance of being selected, using $O(1)$ memory and a single pass over the list.

2.8.1 Solution

Walk over the list and keep one selected element. Select the first element. When you reach the k'th element, replace the selected element with it with probability $\frac{1}{k}$. When the list ends, return the selected element.

Why is this uniform? The k'th element is selected when we reach it with probability $\frac{1}{k}$, and it then has to survive all the following elements: element j

does not replace it with probability $1 - \frac{1}{j} = \frac{j-1}{j}$. So for a list of N elements the chance that the k'th element is returned is

$$\frac{1}{k} \cdot \frac{k}{k+1} \cdot \frac{k+1}{k+2} \cdots \frac{N-1}{N} = \frac{1}{N}$$

which is the same for all the elements.

This is known as reservoir sampling. It also works for selecting m elements: keep the first m, and replace a random one of them with the k'th element with probability $\frac{m}{k}$.

2.9 Two eggs

A tower has 100 floors and you have two identical eggs. You want to find the highest floor from which an egg can be dropped without breaking. An egg that survives a drop can be used again, but a broken egg cannot. What is the smallest number of drops that is enough in the worst case, and how do you conduct the experiment?

2.9.1 Solution

14 drops are enough, and fewer are not.

With one egg there is only one safe method: drop it from floor 1, then 2, then 3 and so on, until it breaks. So with two eggs we use the first egg to find a range of floors, and then check that range one floor at a time with the second egg.

Suppose we allow ourselves d drops. Drop the first egg from floor d. If it breaks, check floors 1 to $d-1$ with the second egg, for a total of at most d drops. If it survives we have used one drop and have $d-1$ left, so the next drop of the first egg can be at most $d-1$ floors higher, at floor $d + (d-1)$. Then $d-2$ floors higher, and so on. So with d drops we can cover at most $d + (d-1) + \cdots + 1 = \frac{d(d+1)}{2}$ floors. For 100 floors we need $\frac{d(d+1)}{2} \geq 100$. Since $\frac{13 \cdot 14}{2} = 91$ is not enough and $\frac{14 \cdot 15}{2} = 105$ is, we need $d = 14$.

So drop the first egg from the floors $14, 27, 39, 50, 60, 69, 77, 84, 90, 95, 99, 100$ until it breaks, and then drop the second egg from each floor above the last floor where the first egg survived, one floor at a time, until it breaks.

2.9.2 Follow up

What if you have three eggs?

2.9.3 Solution

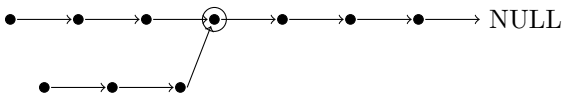
Let $F(e, d)$ be the number of floors that can be handled with e eggs and d drops. The first drop must be from a floor such that, if the egg breaks, the floors below can be handled with $e-1$ eggs and $d-1$ drops, and if it survives, the floors above can be handled with e eggs and $d-1$ drops. So

$$F(e,d) = F(e-1,d-1) + 1 + F(e,d-1)$$

with $F(e,0) = 0$ and $F(0,d) = 0$. This works out to $F(e,d) = \binom{d}{1} + \binom{d}{2} + \cdots + \binom{d}{e}$. For two eggs that is $d + \frac{d(d-1)}{2} = \frac{d(d+1)}{2}$, as before. For three eggs, $F(3,8) = 8 + 28 + 56 = 92$ and $F(3,9) = 9 + 36 + 84 = 129$, so 9 drops are needed.

2.10 Two linked lists that merge

You are given two singly linked lists that merge at some element, from which point on they share the same tail:



Find the first element that is common to both lists (circled in the drawing), in linear time and using $O(1)$ memory.

2.10.1 Solution

Walk over both lists to find their lengths l_1 and l_2. The common tail has the same length in both lists, so the longer list has exactly $|l_1 - l_2|$ extra elements before the merge, compared to the shorter one.

Put a pointer at the start of the shorter list and another pointer $|l_1 - l_2|$ elements into the longer list. Now both pointers are the same distance from the end. Advance them together, one element at a time, until they point to the same element. That element is the first common element. Since both pointers are the same distance from the merge, they reach it at the same time.

This takes $O(l_1 + l_2)$ time and needs only a few variables.

Chapter 3

Combinatorics

3.1 100 urns

There's a room with 100 urns, 2 people standing outside, while they decide on what to do 100 notes with the numbers $1, 2, \ldots, 100$ are randomly placed inside the urns. One of the people from outside will enter the room and will examine all the urns. That person will be allowed to switch at most one pair of notes. That person is then whisked away to a vacation in Mali. The other one will then enter, be given a random number and will be asked to find the note with that number on it. He needs to find it while checking as few urns as possible. Design a strategy for the pair that will allow the second person to look into less than 50 urns to find any number required and in general look at the least number of urns in order to find the note with the required number on it.

3.1.1 Solution

Let's give a solution which is not good, analyze it and then improve it. Here is the bad solution. Whenever the second person comes in he is given the task of finding the note marked x. He will proceed to look into urn x. If x is there he is done. If not, then some other note y is there. He will process to do the same for y.

Analysis: Can the second person get stuck in an infinite loop? No. There are only 100 urns so in the worst case scenario he will check all 100 urns and find the note x in the last one. Now let's take this further. Let's define an equivalence relation on the urns: two urns x and y are equivalent if there exists a route from x to y. This is indeed an equivalence relation since there is always a route from x to x. If there is a route from x to y then there is also a route from y to x since in our case there is only one route to each urn (only one urn carries the note with it's number). Since starting a route from y must eventually lead back to y (routes do not just end), it must pass through x. All this means is that our 100 urns are divided into cycles of equivalence.

Now for the first players job. When the first player comes along he will either see all cycles with $size <= 50$ in which case he need not do anything (unless he wants to break up the largest cycle to reduce the expected value of urns checked by the second player) or will be confronted with utmost a single cycle with $size > 50$ which means he needs to break just this one cycle.

How does one go about "breaking" a cycle? A cycle is a list of urns $x_1, x_2, \ldots, x_{n-1}, x_n$ that make up a cycle, which means their content is $x_2, x_3, \ldots, x_n, x_1$. In order to break the cycle you need to find the center of the cycle. If the length of the cycle is odd then the center is the middle element if the length is even pick any of the central two elements. In our case the longest cycle cannot be of length greater than 100 which means that breaking it up in the center will break it up to pieces less or equal to 50. Let's say the center is in urn k. Switch the content of urns k and n. This will create two cycles: $x_1, x_2, \ldots, x_k$ with content $x_2, x_3, \ldots, x_1$ and another cycle $x_{k+1}, x_{k+2}, \ldots, x_n$ with content $x_{k+2}, x_{k+3}, \ldots, x_{k+1}$.

3.1.2 Follow up

A similar riddle. In this one you do not need to change the content of any urn. Imagine that 100 prisoners numbered zero to a hundred need to find their own number in the urns above and follow the same algorithm as in the solution. What are the chances that they will all succeed? Phrased a different way we ask what are the chances that given that the notes in the urns were randomly distributed what are the chances that there are no cycles greater than 50?

In this version each prisoner may open at most 50 urns, and if even one of them fails to find his number they are all executed. If every prisoner opens urns at random, each one succeeds with probability 1/2 and they all succeed with probability $(1/2)^{100}$, which is practically zero.

3.1.3 Solution to the follow up

Every prisoner x starts at urn x and follows the notes, as in the solution above. He walks along the cycle that contains urn x, and the note x is in the last urn of that cycle before it returns to urn x. So he finds his number within 50 urns exactly when his cycle has at most 50 urns. All the prisoners succeed exactly when there is no cycle longer than 50.

Let's count the arrangements of the notes that have a cycle of length $k > 50$. There can be only one such cycle, since two would need more than 100 urns. Choose the k urns of the cycle in $\binom{100}{k}$ ways, arrange them in a cycle in $(k-1)!$ ways and arrange the other notes in $(100-k)!$ ways. That gives $\binom{100}{k}(k-1)!(100-k)! = \frac{100!}{k}$ arrangements out of 100!, so the chance of a cycle of length k is exactly $\frac{1}{k}$. The chance that they all succeed is

$$1 - \sum_{k=51}^{100} \frac{1}{k} \approx 0.312$$

which is about 31%, an enormous improvement over $(1/2)^{100}$. For $2n$ prisoners

the chance is $1 - \sum_{k=n+1}^{2n} \frac{1}{k}$, which tends to $1 - \ln 2 \approx 0.307$, so it stays above 30% no matter how many prisoners there are.

3.2 Always lead in election

An election was voted a perfect tie (even number of electors) and was decided by way of putting all the votes into a hat, mixing them uniformly and counting them one by one. What is the chance that one of the candidates was leading during the entire counting process?

3.2.1 Solution

The question could be rephrased as how many graphs that go either one step up or one step down (let's call those binary graphs) do not go down below zero out of the set of all graphs that go from zero to zero. At first let's note that in the following solution n is even and so there is no problem with $n/2$. A first observation is that the set of all graphs going from 0 to 0 in n steps is $\binom{n}{n/2}$.

Let's try to count the graphs that tip below 0. Every graph that tips below 0 will reach -1 at one or more points. Let's find the first place where the graph reaches -1. From that point on the graph climbs one more than it descends since it finally reaches 0. Let's flip it from that point on - meaning switch every climb with descent and every descent with a climb. Now the graph descends one more than it climbs and since it starts from that point at height -1 then it now reaches a height -2 at n instead of the original 0.

The main proposition is that *every* graph which travels from 0 to -2 corresponds with a *1-1 correspondence* to graphs that travel from 0 to 0 and reach -1 at some point (convince yourself of this).

If this is so then the number of graphs that go from 0 to -2 is $\binom{n}{n/2+1}$ since we need to choose points at which the graph will go down and we have $n/2+1$ descends. If so then the number of graphs that do not reach -1 is:

$\binom{n}{n/2} - \binom{n}{n/2+1} = \frac{(n)!}{(n/2+1)!(n/2)!} = \frac{1}{n/2+1}\binom{n}{n/2}$.

And now back to the original question "What is the chance that one of the candidates was leading during the entire counting process?". Let's divide the latter result with the former and get:

$\frac{1}{n/2+1}\binom{n}{n/2} / \binom{n}{n/2} = \frac{1}{n/2+1}$

This is the chance that one particular candidate was never behind. "Leading" can only mean "never behind" here, since the count starts and ends in a tie. If a candidate is never behind then the first vote is theirs, and so the other candidate is behind after the first vote: the two events never happen together, and the chance that one of the two candidates was never behind is $\frac{2}{n/2+1}$.

The result is all about Catalan numbers and Bertrand's ballot theorem.

One of the interesting results of this question is that $\binom{n}{n/2}$ is divisible by $n/2+1$. The direct reason is the identity used above, $\binom{n}{n/2} - \binom{n}{n/2+1} = \frac{1}{n/2+1}\binom{n}{n/2}$, whose left side is a difference of two integers. The same flipping

works for any $k \le n/2$. Of the $\binom{n}{k}$ graphs with k steps down, the ones that reach -1 are flipped into graphs which end at $-(n-2k)-2$, and those have $n-k+1$ steps down, so the graphs that never go below 0 number

$$\binom{n}{k} - \binom{n}{k-1} = \frac{n-2k+1}{n-k+1}\binom{n}{k}$$

and so $n-k+1$ always divides $(n-2k+1)\binom{n}{k}$. This is the ballot theorem: if one candidate gets $n-k$ votes and the other k, the chance that the first was never behind during the count is $\frac{n-2k+1}{n-k+1}$. For $k = n/2$ it is the $\frac{1}{n/2+1}$ found above.

3.3 The five card trick

A magician and an assistant perform the following trick. Someone from the audience picks any 5 cards from a standard deck of 52 cards and hands them to the assistant. The assistant looks at them, hides one of them, and puts the other 4 face up on the table in a row, in an order of his choice. The magician, who did not see anything until now, looks at the 4 cards and names the hidden card. There is no other communication between them. How is the trick done?

3.3.1 Solution

There are 5 cards and only 4 suits, so at least two of the cards have the same suit. One of these two is hidden and the other one is put first, which tells the magician the suit of the hidden card.

Now the value. Arrange the 13 values in a circle: $A, 2, 3, \ldots, 10, J, Q, K$ and then A again. Going clockwise from one of the two cards to the other takes some number of steps, and going the other way takes 13 minus that number. One of the two ways is at most 6 steps. The assistant hides the card that is reached at the end of the short way and puts the other card first. So the magician knows that the hidden card is 1 to 6 steps clockwise from the first card. For example, if the two cards are the Q and the 3 of hearts, the assistant hides the 3, which is 4 steps after the Q $(K, A, 2, 3)$.

The remaining 3 cards encode the number of steps. Agree on an order of the whole deck, for example by value and then by suit, so that among the 3 cards there is a low, a middle and a high card. They can be put down in $3! = 6$ orders, one for each number of steps:

low, middle, high	1
low, high, middle	2
middle, low, high	3
middle, high, low	4
high, low, middle	5
high, middle, low	6

The magician reads the suit and the starting value from the first card, the number of steps from the order of the other three, and names the hidden card. This trick is due to the mathematician William Fitch Cheney.

3.4 Frogs over the x axis

Frogs sit on points of the plane with whole coordinates. A frog may jump over a frog that is right next to it (north, south, east or west) and land on the empty point right behind it, at a distance of 2 from where it started. The frog that was jumped over is removed.

At the start there is a frog on every point (x, y) with $y \leq 0$, and all the points with $y > 0$ are empty. How far north can a frog get after a finite number of jumps? Specifically, prove that no frog can reach the line $y = 5$.

3.4.1 Solution

Assume a frog can reach some point on the line $y = 5$; call this point the target. Let $\sigma = \frac{\sqrt{5}-1}{2} \approx 0.618$. It is the positive root of $\sigma^2 + \sigma = 1$. Give a frog that sits at distance d from the target the weight σ^d, where the distance is measured along the grid, $|x - x_0| + |y - y_0|$, and let the weight of a set of frogs be the sum of their weights. A frog on the target has weight 1.

Every jump replaces two frogs by one. There are only three kinds of jumps:

1. A frog at distance $d + 2$ jumps over a frog at distance $d + 1$ and lands at distance d. The weight changes by $\sigma^d - \sigma^{d+1} - \sigma^{d+2} = \sigma^d(1 - \sigma - \sigma^2) = 0$.

2. A frog at distance d jumps over a frog at distance $d + 1$ to distance $d + 2$. The weight changes by $\sigma^{d+2} - \sigma^d - \sigma^{d+1} < 0$.

3. A frog lands at the same distance it started from (for example when jumping across the column of the target). The weight changes by minus the weight of the frog that was jumped over, which is negative.

So the weight of the frogs never grows.

Now let's compute the weight of all the frogs at the start. Put the target at $(0, 5)$. The frogs on the line $y = 0$ are at distances $5, 6, 6, 7, 7, \ldots$ from the target, so their weight is

$$\sigma^5 + 2\sigma^6 + 2\sigma^7 + \ldots = \sigma^5 + \frac{2\sigma^6}{1-\sigma} = \sigma^5 + 2\sigma^4$$

using $1 - \sigma = \sigma^2$. Every line further south is one step further away, so the weight of all the frogs is

$$\frac{\sigma^5 + 2\sigma^4}{1-\sigma} = \sigma^3 + 2\sigma^2 = (2\sigma - 1) + 2(1 - \sigma) = 1$$

using $\sigma^2 = 1 - \sigma$ and $\sigma^3 = \sigma - \sigma^2 = 2\sigma - 1$.

A finite number of jumps involves only a finite number of frogs. Their weight at the start is a finite part of a sum of positive numbers that adds up to 1, so it is strictly less than 1. The jumps never increase it, so at the end it is still less than 1, and no frog can be on the target.

On the other hand, frogs can reach the line $y = 4$ (it takes 20 frogs), so 4 is as far north as they can get. This riddle is known as Conway's soldiers.

3.5 Regrouping coins

This riddle is from Doron.

A collection of $\frac{n(n+1)}{2}$ identical coins is split into groups in some way. Now repeat the following step: take one coin from every group and put all the coins you took into a new group. A group that runs out of coins disappears. Prove that eventually the groups will have the sizes $1, 2, 3, \ldots, n$.

3.5.1 Solution

First notice that the sizes $1, 2, \ldots, n$ do not change once they are reached: taking a coin from each of the n groups leaves groups of sizes $0, 1, \ldots, n-1$ and creates a new group of size n.

Draw the groups as rows of coins, sorted from the largest group at the top to the smallest at the bottom, with all rows starting at the left. The coin in row i and column j sits at position (i, j), and we say it is at level $i + j$. Taking one coin from every group means taking the first column, and the new group becomes a new row. Put the new row at the top for now, and push all the other rows down by one. A coin at (i, j) with $j \geq 2$ moves to $(i + 1, j - 1)$ and a coin of the first column at $(i, 1)$ moves to $(1, i)$. Either way, every coin stays on its level. Then the new row may have to move down to its place to keep the rows sorted.

Let the energy of the coins be the sum of their levels. The step that moves the coins keeps the energy. Moving the new row down to its place lowers the energy: it moves the new row to a larger row number, but it moves the rows it passes, which are longer, to smaller row numbers. So the energy never grows, and since it is a positive whole number it can only go down a finite number of times. From some point on the new row is always at its place at the top, and the coins only move along their levels.

A level $t + 1$ has t positions, $(1, t), (2, t - 1), \ldots, (t, 1)$, and every step moves the coins on it one position along this list, cyclically, with $(t, 1)$ moving to $(1, t)$. So from that point on, the coins on level $t + 1$ simply rotate with a period of t steps.

Suppose the groups never reach $1, 2, \ldots, n$. Then the levels $2, 3, \ldots, n + 1$, which together have room for exactly $\frac{n(n+1)}{2}$ coins, are not all full. Let level $d + 1$ be the lowest one that is not full. The coins that are not on the levels below it are more than it can hold, so there are coins above it, and because the rows start at the left there is a coin on level $d + 2$. The empty position on level $d + 1$ rotates with period d and the coin on level $d + 2$ rotates with period $d + 1$. Since d and $d + 1$ have no common divisor, by the Chinese remainder theorem there is a step at which both are in the same row r: the empty position at $(r, d + 1 - r)$ and the coin right after it at $(r, d + 2 - r)$. But a row cannot have a coin right after an empty position. This contradiction shows that the groups do reach the sizes $1, 2, \ldots, n$.

3.6 Six people

Prove that in every group of 6 people there are either three people who all know each other or three people none of whom knows the others.

In the language of graphs: color every edge of the complete graph on 6 vertices red or blue. Prove that there is a triangle whose edges all have the same color.

3.6.1 Solution

Pick any person A. There are 5 other people, so by the pigeonhole principle A either knows at least 3 of them or does not know at least 3 of them.

Suppose A knows three people B, C and D. If any two of B, C, D know each other then they form, together with A, three people who all know each other. Otherwise B, C and D are three people none of whom knows the others.

If A does not know three people B, C and D the argument is the same with "know" and "don't know" swapped.

The number 6 cannot be lowered: seat 5 people around a table and let everybody know exactly their two neighbors. There are no three people who all know each other and no three people none of whom knows the others.

3.7 Seventeen points and three colors

Every edge of the complete graph on 17 vertices is colored with one of three colors. Prove that there is a triangle whose edges all have the same color.

3.7.1 Solution

This builds on the previous riddle, which shows that if the edges of the complete graph on 6 vertices are colored with two colors then there is a triangle of a single color.

Pick any vertex A. It has 16 edges, each in one of three colors, so by the pigeonhole principle at least 6 of them have the same color, say red. Let $B_1, \ldots, B_6$ be the other ends of these red edges.

If any edge B_iB_j is red then A, B_i, B_j is a red triangle. Otherwise all the edges between $B_1, \ldots, B_6$ are colored with only the two other colors. That is a complete graph on 6 vertices colored with two colors, so it has a triangle of a single color.

Chapter 4

Geometry

4.1 The crescent moon

Prove, in the simplest way you can, that the crescent of the moon is not caused by the shadow of the earth falling on the moon (with the earth standing between the sun and the moon).

4.1.1 Solution

Sometimes, around sunset, you can see both the sun and a crescent moon in the sky at the same time. You are standing on the earth and see both of them in front of you, so the earth is certainly not between the sun and the moon, and still the moon is a crescent.

In fact a crescent moon is always close to the sun in the sky: it shows up only shortly after sunset or shortly before sunrise. The crescent is the half of the moon that faces the sun, seen mostly from the side. The shadow of the earth does fall on the moon sometimes, but that is a lunar eclipse, which can only happen at full moon, when the moon is on the opposite side of the sky from the sun.

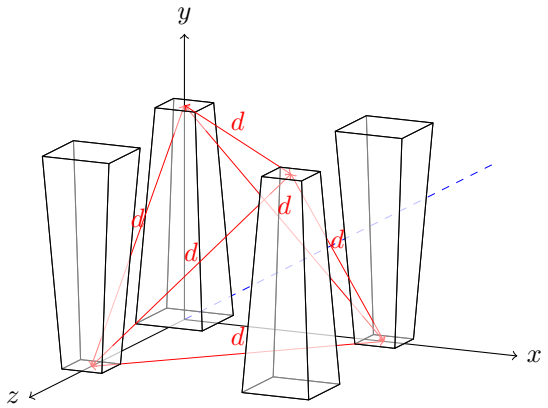
4.2 Four Coke bottles on a table

Arrange four Coke bottles on a table so that the distances between each pair of caps will be the same. Assume that you can make a Coke bottle stand on it's head (this is a pretty strong hint!, maybe I should remove it?!?). The length or height of each bottle is h.

4.2.1 Solution

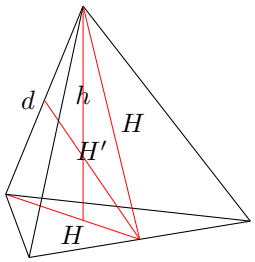
There are two solutions:

The first solution is to put the bottles in a square where the side of the square is the length of the bottle. Bottles next to each other will stand *in reverse* (if the first is on it's head the the second will stand straight).



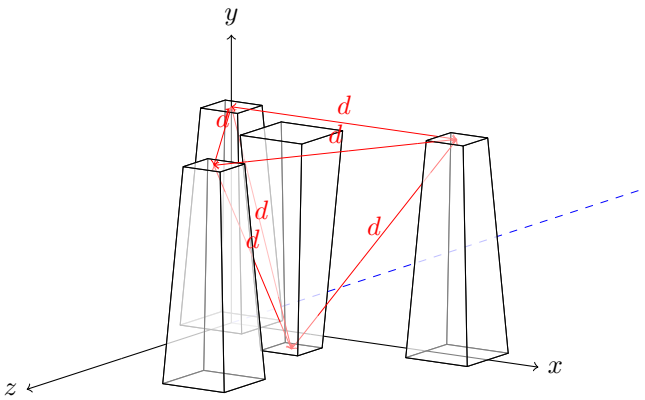
The shape created by the bottle caps is, of course, a perfect triangular pyramid with a side of length $d = \sqrt{2h^2}$.

The second solution is to put three bottles on the vertices of an equilateral triangle standing one way. The fourth will be placed in the middle of the triangle and will be standing reversed to the others. The problem here is to calculate the side of the triangle. Let's use the following diagram:



What we are trying to find is d as a function of h. Since H, $d/2$ and d create a right angled triangle, it follows that $H^2 + (d/2)^2 = d^2$ and so $H = \sqrt{3}/2d$. Since the size of the inner triangle could be computed two ways then $hH = dH'$ and since $H' = \sqrt{H^2 - (d/2)^2}$ it follows after some math that $d = \sqrt{3/2}h$ which makes d about $1.224h$.

The solution built on this analysis will look like this:



As is evident the key to solving this question is the perfect triangular pyramid as both solutions are based on it and it is the only way to put 4 elements in space being spaced equally apart. In theory you could place the bottles at any height and so infinite solutions to this riddle could be produced by making the side of the perfect pyramid as small or as large as you want. But we are constrained by the table and so the heights at which the caps are will be either 0 or h. The two solutions above seem to be primary examples under this constraint. Minor variations on the above two solutions could be produced by making the bottles whose caps are at table level lay instead of stand on their heads. The first solution is more attractive than the second because it could be performed in practice without relying on measurement tools in order to place the bottles at the calculated distance from each other as per the second solution.

4.3 Jumping frogs

Four frogs are sitting on the vertices of a square the side of which is of length 1. In turns, a frog may leap over any other frog jumping twice the distance between them. Could the frogs, after any number of steps, form a square of a size different than 1? Another version would be: can you prove that if, after any number of jumps, the frogs form a square then the size of the square will be 1?

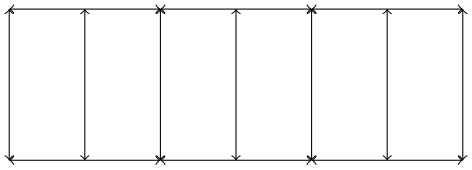
4.3.1 Solution

If we put the frogs on an infinite grid of size 1 then after each jump all frogs must be located on that grid. So if the frogs can create a square of a size bigger than 1 then that size must be a whole number which is bigger than 1. Now let's note that if the frogs can create a square bigger than 1 then, starting from that square, then can reverse their steps and create a square which is of size 1. This means that the frogs can make the square they form bigger or smaller by some factor at any time. The problem with this is that this is also true for the

original sized 1 square: the frogs can make it smaller by that same factor but that is a contradiction to the fact that the frogs always remain on the grid.

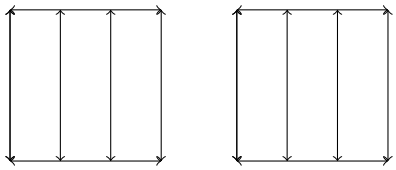
4.4 Pigsty for six pigs

A farmer has obtained 13 panels in order to create 6 identical pigsties. He was planning to arrange them like this:



Unfortunately, one of his panels got stolen and he now only has 12 panels. Can you design 6 identical pigsties with 12 panels?

4.4.1 Solution



4.5 A shape avoiding whole points

A shape with area less than 1 is drawn on the plane. Show that the shape can be moved (without rotating it) so that it does not contain any point (x, y) in which both x and y are whole numbers.

4.5.1 Solution

Cut the plane, together with the shape, along the lines $x = m$ and $y = n$ for all whole numbers m and n, into unit squares. Now slide all the unit squares on top of the square $[0, 1) \times [0, 1)$, as if they were transparent slides stacked on each other, without rotating them. Each slide carries the piece of the shape that was in it.

The pieces have a total area of less than 1, so even stacked together they cannot cover the whole unit square: there is a point (a, b) in the unit square that is not covered by any piece.

Move the shape by $-a$ along the x axis and by $-b$ along the y axis. A whole point (m, n) is in the moved shape exactly when the point $(m + a, n + b)$ is in the original shape. That point lies in the slide of the unit square whose corner is (m, n), at the place that lands on (a, b) when the slides are stacked. But no

piece covers (a, b), so $(m + a, n + b)$ is not in the shape, and the moved shape contains no whole point.

4.6 Strange Trip

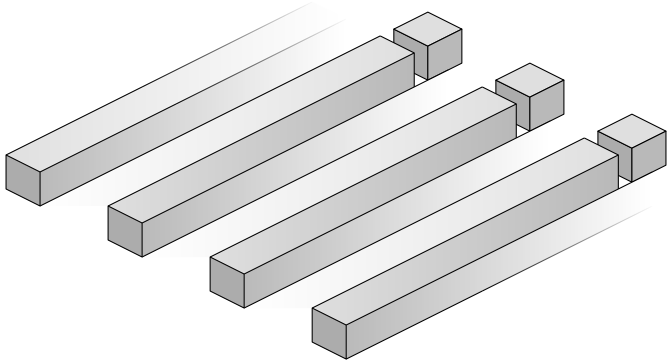
You travel 100 miles north, 100 miles east, and then 100 miles south. You are at the same point that you started from. Describe all the places on earth this could be, if any.

4.6.1 Solution

At the south pole. Or, 100 miles south of any point on the circle around the north pole that is 100 miles in circumference. Or, 100 miles south of any such circle whose circumference is an integral fraction of 100 miles.

4.7 Three or four boxes

Are there three or four boxes in the following drawing? How do you explain the illusion?



4.7.1 Solution

The heart of the illusion is the gradual transformation of floors on the right hand side of the drawing into sides of boxes in the left hand side of the drawing. That is also the reason for the gradual transformation of color of the floors.

Chapter 5

Information theory

5.1 Coins on a chess board

On a standard 8 by 8 chess board lie 64 coins, one on each square, each showing either heads or tails. Player A enters the room. A host points at one of the squares. Player A must flip exactly one coin of his choice and is then whisked away to a vacation in the Maldives. Player B enters the room and must point at the square that the host chose.

Find a strategy that A and B can agree on before the game that always succeeds.

5.1.1 Solution

Number the squares 0 to 63 and think of each number as 6 bits. Define the value of the board as the bitwise exclusive or (xor) of the numbers of all the squares whose coin shows heads. Player B simply computes the value of the board and points at that square.

So the job of player A is to make the value of the board equal to the square t that the host chose. Let v be the value of the board when A enters. Flipping the coin on square k either adds k to the set of heads or removes it, and in both cases the value of the board becomes $v \oplus k$. So A flips the coin on square $k = v \oplus t$ and the value becomes $v \oplus v \oplus t = t$. If the value of the board is already t then $k = 0$, and flipping the coin on square 0 does not change the value.

This works because flipping one of 64 coins gives A exactly 64 choices, one for each value of the board.

5.2 The poisoned glass

A king has 1000 glasses of wine and knows that exactly one of them is poisoned. The poison is so strong that a single sip kills, but it acts slowly: whoever drinks it dies a day later. The king wants to know which glass is poisoned by tomorrow,

and he has 10 servants who can taste the wine. Each servant may take a sip from as many glasses as he likes. How can the king find the poisoned glass?

5.2.1 Solution

Number the glasses 0 to 999 and write each number in binary using 10 bits. That is enough, since $2^{10} = 1024 \geq 1000$. Number the servants 0 to 9. Servant i sips from every glass whose number has a 1 in bit i.

The next day, look at which servants died. Servant i died exactly when bit i of the number of the poisoned glass is 1. So writing a 1 for every servant that died and a 0 for every servant that survived spells out the number of the poisoned glass in binary. If nobody dies, the poisoned glass is glass number 0.

10 servants is the least possible: each servant either dies or not, so 9 servants can produce only $2^9 = 512$ different outcomes, which is not enough to tell apart 1000 glasses.

5.3 The spy

A spy in an enemy country wants to send information home using the enemy's daily morning radio transmission, which is 15 bits long. Every morning the transmission is random. The spy can get into the radio station a few minutes before the transmission, look at the 15 bits that are about to be sent, and either leave them as they are or flip a single bit (flipping more would make the transmission look corrupt). His operators listen to the transmission and decode it according to a method agreed on in advance. How much information can the spy send every day?

5.3.1 Solution

The spy can send 4 bits, that is, one of 16 messages, every day.

He cannot do better: every morning he has exactly 16 choices (flip one of the 15 bits, or flip nothing), and the operators see only the result, so at most 16 different messages can be told apart.

Here is how to reach 16. Number the positions of the bits 1 to 15 and think of each position number as 4 bits. The operators decode the transmission by computing the exclusive or (xor) of the positions of all the bits that are 1. That is a number between 0 and 15, and it is the message.

The spy computes the same value v for the transmission he finds. He wants to send the message m. Flipping the bit at position k adds k to the set of positions of the 1 bits or removes it, and either way the value becomes $v \oplus k$. So if $v = m$ he leaves the transmission alone, and otherwise he flips the bit at position $k = v \oplus m$, which is between 1 and 15. The value becomes $v \oplus v \oplus m = m$.

This is the Hamming code of length 15 in disguise. Compare with the riddle of the coins on the chess board, which uses the same trick.

5.3.2 Follow up

Two prisoners agree on a method in advance. Then the first is told a two digit number (one of 00 to 99) and taken into a room with 11 dials, each of which has 10 positions and is set at random. He must turn exactly one dial to a different position. Then the second prisoner enters, looks at the dials and has to say the number. Can they do it?

5.3.3 Solution to the follow up

No. The first prisoner has only $11 \cdot 9 = 99$ different moves, and the second sees only the result, so at most 99 numbers can be told apart, one short of the 100 needed.

What if he may also leave the dials alone? Then he has exactly $1 + 99 = 100$ choices, so from *every* setting of the dials the 100 settings he can reach must all decode to different numbers. That makes the settings that decode to any one number a perfect code: every setting of the dials is at most one move away from exactly one of them. With a prime number q of positions per dial (or a power of a prime) and $q + 1$ dials this works, the same way as the spy's 15 bits, with arithmetic modulo q in place of xor: for example 8 dials with 7 positions can carry one of 49 numbers. But 10 is not a prime power, and whether a perfect code of this kind exists for 10 positions is, as far as I know, an open problem.

5.4 Twelve coins

You have 12 coins that look the same. One of them is fake and has a different weight from the others, but you do not know whether it is heavier or lighter. Using a balance scale only 3 times, find the fake coin and tell whether it is heavier or lighter. There are two different kinds of solution: one in which every weighing depends on the results of the previous ones, and one in which all three weighings are decided in advance. Find both.

5.4.1 Solution

First, why this is tight: there are 24 possibilities (12 coins, each heavier or lighter) and every weighing has 3 results (left side heavier, right side heavier, or balance), so 3 weighings have 27 results, just enough. This means every weighing must split the remaining possibilities as evenly as possible.

Adaptive solution. Number the coins 1 to 12. First weigh $1, 2, 3, 4$ against $5, 6, 7, 8$.

If they balance, the fake coin is one of $9, 10, 11, 12$ and the coins 1 to 8 are genuine. Weigh $9, 10, 11$ against $1, 2, 3$. If they balance, coin 12 is fake; weigh it against coin 1 to see whether it is heavier or lighter. Otherwise we know whether the fake coin, one of $9, 10, 11$, is heavier or lighter, and weighing 9 against 10 finds it: the one that tips the scale the known way is fake, and if they balance it is 11.

If the left side is heavier, then either one of $1, 2, 3, 4$ is heavy or one of $5, 6, 7, 8$ is light, and 9 to 12 are genuine. Weigh $1, 2, 5$ against $3, 6, 9$. If they balance, the fake is 4 (heavy), 7 or 8 (light): weigh 7 against 8, the lighter one is fake, and if they balance 4 is the heavy fake. If the left side is heavier, the fake is 1 or 2 (heavy) or 6 (light): weigh 1 against 2, the heavier one is fake, and if they balance 6 is the light fake. If the right side is heavier, the fake is 3 (heavy) or 5 (light): weigh 3 against the genuine 9. If 3 is heavier it is the fake, and if they balance 5 is the light fake.

If the right side is heavier the first time, do the same with the roles of heavy and light swapped.

Non adaptive solution. Do these three weighings:

weighing	left side	right side
1	4, 5, 6, 8	1, 2, 3, 7
2	7, 9, 10, 11	1, 2, 6, 8
3	2, 6, 9, 12	3, 5, 8, 11

Give every coin a code of three symbols, one per weighing: L if the coin is on the left side, R if it is on the right side, and $-$ if it is not on the scale. For example coin 6 has the code LRL and coin 12 has the code $--L$. Record the result of each weighing the same way: L if the left side was heavier, R if the right side was heavier and $-$ if they balanced.

If the fake coin is heavy, the result is exactly its code. If it is light, the result is its code with L and R swapped. The codes were chosen so that no two coins have the same code or swapped codes, so the result points at exactly one coin and tells whether it is heavy or light. For example the result RLR is the code of coin 6 swapped, so coin 6 is light. Every weighing has 4 coins on each side, and there are 26 possible codes besides $---$, which come in 13 swapped pairs, of which we use 12.

5.5 Thirteen coins

The same as the previous riddle, but with 13 coins: one of them is fake and is either heavier or lighter than the others. With 3 weighings, can you find the fake coin and tell whether it is heavier or lighter? If not, what can you do?

5.5.1 Solution

Counting alone does not rule it out: there are 26 possibilities and 27 results of three weighings. But it cannot be done.

Suppose the first weighing puts k coins on each side. If the scale balances, the fake coin is one of the other $13 - 2k$ coins, and there are $2(13 - 2k)$ possibilities left for only 9 results of the two remaining weighings, so $26 - 4k \leq 9$, which means $k \geq 5$. If the left side is heavier, the fake is one of the k coins on the left and heavy, or one of the k coins on the right and light. That is $2k$ possibilities

for 9 results, so $k \leq 4$. Both cases can happen, so we need both $k \geq 5$ and $k \leq 4$, which is impossible.

What can be done is to always find the fake coin, and almost always tell whether it is heavier or lighter. Weigh $1, 2, 3, 4$ against $5, 6, 7, 8$. If the scale does not balance, coins 9 to 13 are genuine and we go on exactly as in the twelve coin riddle, finding the coin and its weight. If it balances, the fake coin is one of 9 to 13 and coins 1 to 8 are genuine. Weigh $9, 10, 11$ against $1, 2, 3$. If this does not balance we know whether the fake is heavy or light, and weighing 9 against 10 finds it. If it balances, the fake is 12 or 13. Weigh 12 against 1: if it does not balance then 12 is the fake and we see whether it is heavy or light, and if it balances then 13 is the fake, but we never weighed it, so we do not know whether it is heavier or lighter.

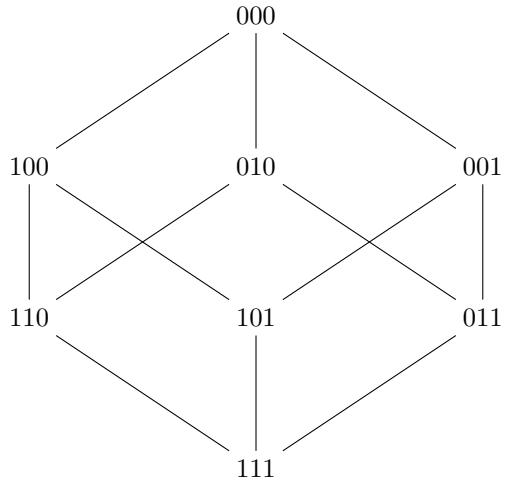
If we are also given one extra coin that is known to be genuine, then 13 coins can be solved completely, since the genuine coin lets the first weighing put 5 unknown coins against 4 unknown coins and the genuine coin.

5.6 Two bits out of three using one

You need to setup a communication protocol to transport 3 bits at a time. Whenever you are given three bits of information you may only pass one bit over the medium to represent that information. The receiver should be able to produce the 3 bits with a maximum of 1 bit of mistake (anywhere).

5.6.1 Solution

If we put all 8 possible 3 bit combinations in a graph where two combinations are connected via an edge iff there is just 1 bit of information difference between them we see that 000 and 111 are at the edges of the graph and are 1 bit close to $001, 010, 100$ and $011, 101110$ respectively. This means that the entire group of 8 possibilities could be divided into 2 groups of 4 thus: $000, 001, 010, 100$ and $111, 011, 101, 110$. So when we get 3 bits of information we will send 0 for any member of the first group and 1 for any member of the second. The receiver will guess 000 for 0 and 111 for 1. This means that the receiver will be wrong by at most 1 bit. The chance that the receiver will get the 3rd bit right is 1 in 4 if the input is evenly distributed. We can also reverse the meaning of 0 and 1 and have the decoder guess 111 for 0 and 000 for 1. Actually we are sending 0 when most bits are 0 and 1 when most bits are 1 and since most out of 3 is at least 2 we always get 2 bits right. Note that we can't choose other representatives for the groups, and not even representatives which are reverse of each other (like 001 and 110) since these are not distanced by at most 1 bit from the rest of the group that they are in (for instance 001 has distance 2 from 100).



If we look at the graph then we see that any bi-partition of the graph would work as well. For instance we could divide the graph to $001, 000, 101, 011$ and $110, 100, 010, 111$. The first group the representative will be 001 and the second group representative will be 110. Actually the graph is so symmetric that any two negative bitwise strings would do to partition it so there are 4 possible protocols (8 if reverse counts too).

Chapter 6

Logic

6.1 Balls on a circular track

An unknown number of balls race around a circular track, each at a constant speed of one lap per second. Some of them go clockwise and some counter clockwise. When two balls collide they immediately reverse direction and continue at the same speed. During ten seconds there were 860 collisions. How many balls are there?

6.1.1 Solution

First let's get rid of the change of direction. The balls are identical, so a collision in which two balls bounce off each other looks exactly the same as two balls passing through each other. So we can pretend that the balls never change direction and just pass each other, and the number of collisions stays the same.

Now every ball keeps its direction forever, and after one second every ball is back where it started. So the balls do exactly the same thing every second and the number of collisions per second is constant: $860/10 = 86$.

Let X be the number of balls going clockwise and Y the number going counter clockwise. A clockwise ball and a counter clockwise ball approach each other at two laps per second, so they meet twice every second. Two balls going in the same direction never meet. So there are $2XY$ collisions per second, which gives $2XY = 86$ or $XY = 43$. Since 43 is prime, one direction has a single ball and the other has 43 balls. In total there are 44 balls.

6.2 Blue and red martians

On mars live two types of aliens: ones with blue dots on their head and ones with red dots. No martian knows what color dot he/she has on their head and they do not have verbal skills. One day they get told that a new virus is out there that only kills the blue kind and therefor all the blue martians must leave

the planet to space where the virus cannot survive. Red martians, however, must not leave the planet since they cannot survive in outer space. Each day a ship, large enough to hold all the martians, will leave at noon carrying martians to outer space. Can just the blue martians leave? if so how? P.S. there is at least one martian with a blue dot on his head and everyone knows so.

6.2.1 Solution

Since the martians are very smart, if there are N martians then all of them will leave on day N. How is that? The proof is by induction. Imagine that there is only one martian with a blue dot. He sees everyone with red dots on their head and since he knows that there is at least one martian with a blue dot on his/her head he deduces that it is he. He therefor leaves in the first possible occasion, meaning the first day. Now imagine that the proposition is true for N. Now imagine that there are $N+1$ martians. Each of them sees only N blue dots in the crowd. The deduce that there are either N martians with blue dots (excluding themselves) or $N+1$ blue dotted martians (including themselves). They don't know which. So they wait for day N. If there were indeed N martians all of them would have left on day N. They do not leave. From this each one of the $N+1$ deduces that the other N martians see N blue dots in the crowd which means that they themselves are blue dotted which means that there are $N+1$ blue dotted martians which include themselves. This prompts them to leave on the first possible occasion, which is day $N+1$.

6.3 Five pirates

Five pirates find a sack with 100 gold coins and want to divide them. The pirates are ranked by seniority: the most senior is #5, then #4 and so on down to the most junior, #1. The most senior pirate proposes how to divide the coins and then all the pirates, including him, vote on the proposal. If at least half of the votes accept it, the coins are divided accordingly and they are done. Otherwise the pirate who made the proposal is shot, and the next in line makes a proposal, and so on.

Every pirate is perfectly rational. His first priority is to stay alive, his second is to get as many coins as possible, and his third is blood: if a vote makes no difference to him otherwise, he votes against, hoping for a kill.

What should pirate #5 propose to stay alive and keep as many coins as possible?

6.3.1 Solution

Pirate #5 gives one coin to pirate #3 and one coin to pirate #1 and keeps the other 98.

To see why, solve the problem backwards, starting with fewer pirates.

Two pirates. #2 proposes to keep all 100 coins. He votes for it, which is already half of the votes, so it passes. #1 gets nothing.

Three pirates. #3 needs two votes. He knows that if he is shot, #1 gets nothing. So he offers #1 a single coin, which is better for #1 than what he would get otherwise, and #1 votes for it. Proposal: #3 gets 99, #2 gets 0, #1 gets 1.

Four pirates. #4 needs two votes out of four, his own and one more. If he is shot, #2 gets nothing, so a single coin buys his vote. Proposal: #4 gets 99, #3 gets 0, #2 gets 1, #1 gets 0.

Five pirates. #5 needs three votes out of five, his own and two more. If he is shot, #3 and #1 get nothing, so a single coin buys each of their votes. Proposal: #5 gets 98, #4 gets 0, #3 gets 1, #2 gets 0, #1 gets 1.

Note that giving a pirate exactly what he would get anyway is not enough to buy his vote, since he would then vote against for the blood.

6.4 Infinitely many dwarves

A countably infinite number of dwarves stand in a line, numbered $1, 2, 3, \ldots$, all facing the same direction, so that every dwarf sees all the dwarves in front of him (dwarf 1 sees dwarves $2, 3, \ldots$) but not himself or the dwarves behind him. A black or white hat is placed on the head of each dwarf at random. Starting with dwarf 1, each dwarf in turn has to guess the color of his own hat. If he is right he lives, otherwise he is killed. The dwarves can agree on a strategy beforehand.

Is there a strategy in which only a finite number of dwarves guess wrong? Can it be done if the dwarves do not hear each other's guesses (in other words, if they all guess at the same time)?

6.4.1 Solution

Surprisingly the answer to both questions is yes, but the strategy uses the axiom of choice.

Think of an assignment of hats as an infinite sequence of colors. Call two sequences equivalent if they differ in only a finite number of places. This is an equivalence relation, so it splits all the possible sequences into classes. Before the game, the dwarves use the axiom of choice to pick one representative sequence from every class and agree on these representatives.

During the game every dwarf sees all the hats in front of him. That is all of the sequence except for a finite beginning, and changing a finite beginning does not change the class. So every dwarf knows the class of the real sequence, and therefore knows its representative r. Every dwarf k guesses the color r_k.

The real sequence and r are in the same class, so they differ in only a finite number of places, which means that only a finite number of dwarves guess wrong. This strategy does not use the guesses of the other dwarves at all, so it works when they all guess at the same time.

When the dwarves do hear each other they can do even better: everybody except dwarf 1 survives. Dwarf 1 counts the places $k \geq 2$ in which the hats he sees differ from r. There are finitely many of them, and he announces "black" if their number is even and "white" if it is odd. Dwarf k knows his own color from this parity: he sees the hats of the dwarves in front of him, and he heard the correct colors of the dwarves $2, \ldots, k-1$ behind him, so he knows all the differences from r except his own.

Note that this strategy cannot be written down explicitly, since it depends on the axiom of choice. The set of hat assignments in which a given dwarf survives need not be measurable, so asking what the chance of a given dwarf to survive is may have no meaning.

6.5 Liars and Truth tellers

In a circle there are n people, some tall and some short. Tall people always tell the truth while short people always lie. All of them tell you that the two people on their side are of different heights. How many liars and truth tellers are there?

6.5.1 Solution

Let's see what happens on the circle itself. Pick any spot and look at the two people from that spot onward. There could be only 4 possibilities: $(t,t)(l,t)(t,l)(l,l)$. If possibility one occurs then the next person must be a liar for the truth teller in the middle to be right, creating (t,t,l). If possibility two occurs then the next person must be a truth teller for the same reason and we get (l,t,t). If the third option is the one then the next person must be a truth teller so that the liar reports, wrongly, that two people of different height are near him. This will give us (t,l,t). Consider these three options, they all combine to form an endless chain of $(t,t,l,t,t,l,t,t,l,\ldots)$ which means that a third of people are liars. Option 4, (l,l) means that the next person is also a liar and so on forever which gives us all liars. To sum up: either a third of people are liars or all of them.

There is also a formula. Write $a_i = 1$ if person i is tall and $a_i = 0$ if short. A tall person tells the truth, so their two neighbors differ: $a_{i-1}+a_{i+1} \equiv 1 \pmod 2$. A short person lies, so their two neighbors are the same: $a_{i-1} + a_{i+1} \equiv 0 \pmod 2$. In both cases $a_{i-1} + a_{i+1} \equiv a_i$, or $a_{i+1} \equiv a_i + a_{i-1} \pmod 2$, which is the Fibonacci rule taken modulo 2. Whatever two neighbors we start from, the sequence is either $0,0,0,\ldots$ or repeats $0,1,1,0,1,1,\ldots$ with period 3. Going around the circle it must come back to where it started after n steps, so the second pattern needs n to be divisible by 3. This also shows what the solution above glossed over: if n is divisible by 3 then either everybody lies or exactly $n/3$ people are liars, and otherwise everybody lies.

6.6 Reduction in force

A boss tells his 100 employees that tomorrow there will be a reduction in force. He will stand them in a line, one behind the other, all facing the same direction, so that each of them sees everyone in front of them but nobody behind them. He will put a white or a black cap on the head of each of them. Then each one in turn will say out loud the color of their own cap. Whoever is right keeps their job and whoever is wrong is fired. The employees may decide in advance in which order they speak, and agree on a strategy. How many of them can be sure to keep their job?

6.6.1 Solution

99 of them can be sure to keep their job, and the last one has a 50% chance.

Think of white as 0 and black as 1. The employees speak from the back of the line to the front. The one at the back sees all the other 99 caps. She adds them up and says the color that makes the sum of all 100 numbers even: "white" if the sum she sees is even and "black" if it is odd. Her own cap has nothing to do with it, so she is right with probability 1/2.

Now everyone knows that the sum of the 99 caps in front of her is even if she said "white" and odd if she said "black". Each of the following employees knows the color of every one of those 99 caps except their own: they see the caps in front of them, and every employee who spoke before them, except the one at the back, said their color correctly. So they can compute their own color as the one that gives the known parity, and they are all right.

6.7 Three buckets with false info

You have three buckets in front of you. One has only black balls in it. One has only white balls in it. The last one has an even mix of black and white. You don't know which is which. Each bucket carries a note. The notes say "black only", "white only" and "mixed". It is known that all three notes are wrong. You need to design an experiment in which you will extract balls from the buckets of your choosing in order to find out which bucket has which content. Your aim is to minimize the number of ball extractions to attain your goal. How many balls will you need to extract in order to ascertain which content is in which bucket?

6.7.1 Solution

Only one ball needs to be extracted from the bucket which is labeled *Mixed*. Since the bucket is marked as *Mixed* and that is known to be false then the bucket must be either all black or all white. So the single ball we extract tells us which of the two it is. Let's call this color N and it's opposite M. Now we know what this bucket is of color N. One of the two other buckets claims to be of

color M. Since it's label is false it must either be N or the true $Mixed$ bucket. We already know which bucket is N and it's not that one. That means that the bucket labeled as M is actually the true $Mixed$ bucket. The last bucket is the bucket of true M.

6.8 The Wardens challenge

A warden summons all of his prisoners to a talk. "From this day on you will not see each other. I will select, by random and with possible repetitions, one of you each day and will place him or her in a special holding chamber. The holding chamber has a light bulb with a switch. When the day is done I will lead the prisoner back to his or her cell. At any time, any one of who understands that all of you have been in the special holding cell may call out and stop the experiment. If that person is right then you will all go free and if he is wrong you will stay here for life". The prisoners then have time to confer and plan a strategy before they are taken to their cells and are prevented from speaking to one another for the remainder of the experiment. Plan the prisoners strategy.

6.8.1 Solution

A first solution would be to use a special designated person to act as counter. The counter never flips the light switch on. Every time the counter is called into the holding cell and sees the light he/she adds 1 to a sum he/she is storing in memory as to the number of people that have been in the holding cell and flips the light off. Every other prisoner will follow the following rules: If you have never flipped the switch on and the switch is off when you enter the room then switch it on. Otherwise do nothing. The counter is the only person to proclaim that everyone has been to the holding cell and obviously has to count himself/herself as well.

The problem with this solution is that it's efficiency is quite low. Time is also wasted when a prisoner that has never been to the cell enters it and sees the light on which prevents him from turning the light on himself and makes him wait for the next time he/she is called to the holding cell.

Chapter 7

Number theory

7.1 $2^{2n} - 1$

Prove that for each $n \in \mathbb{N} > 0$, $2^{2n} - 1$ is divisible by 3

7.1.1 Solution

$2^{2n} - 1 = (2^n)^2 - 1^2 = (2^n - 1)(2^n + 1)$. 2^n is not divisible by three (because it's primary decomposition is made up of only 2's...). For each three consecutive numbers one must be divisible by 3 and this applies to the following three consecutive numbers: $2^n - 1, 2^n, 2^n + 1$. Since 2^n is not divisible by 3 then either $2^n - 1$ or $2^n + 1$ must be divisible by three and hence, using the equation above, $2^{2n} - 1$ is divisible by 3.

7.2 100 consecutive non primes

Find 100 consecutive numbers none of which is prime.

7.2.1 Solution

Take the numbers $101! + 2, 101! + 3, \ldots, 101! + 101$. These are exactly 100 consecutive numbers ($2, 3, \ldots, 101$ is a list of 100 numbers).

For every k with $2 \leq k \leq 101$, k divides $101!$ and k divides itself, so k divides $101! + k$. Since $101! + k$ is larger than k, it has a divisor other than 1 and itself and is therefore not prime.

The same trick works for any N: the numbers $(N+1)!+2, \ldots, (N+1)!+N+1$ are N consecutive numbers none of which is prime. This shows that there are arbitrarily long gaps between consecutive primes.

7.3 13 rocks

You are given 13 rocks. If you remove any one of them, the remaining 12 can be split into two groups of 6 rocks each that balance each other on a scale. Prove that all 13 rocks have the same weight.

7.3.1 Solution

First assume that all weights $w_1, \ldots, w_{13}$ are whole numbers and let S be their sum. If we remove rock i, the remaining 12 rocks are split into two groups of equal weight, so $S - w_i$ is even. This holds for every i, so all the w_i have the same parity as S: they are either all even or all odd.

Notice that two operations keep the property of the rocks (any 12 balance as 6 against 6): subtracting the same number from all weights (each side of the scale holds 6 rocks, so both sides lose the same amount) and dividing all weights by the same number.

Now subtract the smallest weight from all weights. The property still holds and one of the weights is 0. Since all weights have the same parity as that 0, they are all even. Divide all of them by 2. They are still whole numbers, the property still holds, so they are again all even, and we can divide by 2 again. We can repeat this forever, which is only possible if all the weights are 0. That means that before subtracting, all the weights were equal to the smallest weight.

If the weights are rational, multiply all of them by a common denominator. That gives whole numbers with the same property, which are therefore all equal.

If the weights are any real numbers the argument needs one more step. Each balancing is a linear equation in $w_1, \ldots, w_{13}$ whose coefficients are 0, 1 and -1. The argument above shows that the only rational solutions of this system of equations are of the form $(c, c, \ldots, c)$, so the solution space over the rationals has dimension 1. The dimension of the solution space of a system of linear equations is 13 minus the rank of the matrix, and the rank is computed by Gaussian elimination, which never leaves the rationals when the coefficients are rational. So the rank is the same over the reals and the solution space over the reals also has dimension 1: all the weights are equal.

7.4 Rational points on a circle

Could there be a circle on the plane where only one point has rational co-ordinates? A point $p = (x, y)$ has rational co-ordinates iff $x, y \in \mathbb{Q}$. What about a sphere? Are there circles with more than one rational point? Are there circles with infinite rational points? Can you find a circle with exactly n rational points on it for each $n \in \mathbb{N}$?

7.4.1 Solution

Yes, there could. Consider the circle: $(x - r)^2 + y^2 = r^2$ for some irrational r. $(0, 0)$ is a point on this circle and is rational. But from the equation it follows

that: $x^2 - 2xr + r^2 + y^2 = r^2$ or $x^2 + y^2 = 2xr$ or $r = (x^2 + y^2)/2x$. If there was a rational solution to this equation where $x \neq 0$ then it would follow that r is rational since it is the result of multiplication, division, addition and subtraction of rational numbers and this is a contradiction. This means this equation can only have rational solutions for $x = 0$. If $x = 0$ then $r^2 + y^2 = r^2$ and so $y = 0$ and so $(0, 0)$ is the only such solution.

Same solution applies to a sphere: $(x - r)^2 + y^2 + z^2 = r^2$ for some irrational r. In this case: $r = (x^2 + y^2 + z^2)/2x$. Which means that for $x \neq 0$ any rational solution would imply a rational r. $(0, 0, 0)$ is therefore the only rational solution.

Yes, there are circles with more than one rational point, and as soon as a circle has three rational points it has infinitely many. Three points which are not on one line determine a circle. Its center is where the perpendicular bisectors of two of the chords meet, and since these are lines with rational coefficients the center (a, b) is rational, and so is r^2, the squared distance from the center to one of the points. Now take one rational point $P = (p, q)$ on the circle and draw through it the line $y - q = t(x - p)$ for some rational slope t. Substituting it into $(x - a)^2 + (y - b)^2 = r^2$ gives a quadratic equation in x with rational coefficients. One of its roots is p, and the sum of its two roots is rational (it is minus the ratio of two of the coefficients), so the other root is rational too, and so is the second point where the line meets the circle. Different slopes give different points, so there are infinitely many. For the unit circle, starting from $P = (-1, 0)$, this gives the points $\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$ for $t \in \mathbb{Q}$, for example $(3/5, 4/5)$ for $t = 1/2$. And the circle $x^2 + y^2 = 2$, which has the rational points $(1, 1), (-1, 1), (1, -1), (-1, -1)$, also has infinitely many, for example $(7/5, 1/5)$.

So a circle has 0, 1, 2 or infinitely many rational points, and for $n \geq 3$ there is no circle with exactly n of them. The other three counts all happen. 1 was shown above. For 0 take the circle $(x - \sqrt{2})^2 + y^2 = 1$: a rational point on it would satisfy $x^2 + y^2 + 1 = 2\sqrt{2}x$, which for $x \neq 0$ would make $\sqrt{2}$ rational, and for $x = 0$ gives $y^2 = -1$. For 2 take the circle with center $(1/2, \sqrt{2})$ through $(0, 0)$ and $(1, 0)$, that is $(x - 1/2)^2 + (y - \sqrt{2})^2 = 9/4$: a rational point on it would satisfy $x^2 - x + y^2 = 2\sqrt{2}y$, which for $y \neq 0$ would make $\sqrt{2}$ rational, and for $y = 0$ leaves only $x = 0$ and $x = 1$.

Chapter 8

Probability

8.1 Birthing girls

A village elder wants more women in the village. He tells the villagers: every family should keep having babies until it has a daughter, and then stop. Will this make the village have more girls than boys?

8.1.1 Solution

No. The ratio of boys to girls will stay 1 : 1.

The short explanation: every birth is a boy or a girl with probability 1/2, independently of all previous births. The rule only decides when a family stops, it cannot change the outcome of any single birth. So on average half of all the babies born are girls, no matter what the rule for stopping is.

Here is another way to see it. Let the villagers have as many babies as they like, until every family has a daughter. Here the ratio is clearly 1 : 1. Now remove every child that has an older sister. Each removed child is a boy or a girl with probability 1/2, so the removed children are also 1 : 1, and so are the children that are left. But the children that are left are exactly what the elder's rule produces.

And here is the calculation. Every family has exactly one girl. A family has k boys if its first k babies are boys and the next one is a girl, which happens with probability $(1/2)^{k+1}$. So the expected number of boys in a family is

$$s = \sum_{k=1}^{\infty} \frac{k}{2^{k+1}} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots$$

Subtracting $\frac{s}{2} = \frac{1}{8} + \frac{2}{16} + \dots$ gives

$$\frac{s}{2} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots = \frac{1}{2}$$

so $s = 1$: on average every family has one boy and one girl.

8.1.2 Follow up

What if the elder had said "stop when you have six girls for every boy"? Unlike the previous rule, this condition may never happen. What are the chances that a family will ever stop?

8.1.3 Solution to the follow up

About 5.7%. Almost every family goes on forever.

Keep track of $x = \text{girls} - 6 \cdot \text{boys}$. It starts at 0, every girl adds 1 and every boy subtracts 6, and the family stops when x is back at 0. On average a birth changes x by $\frac{1-6}{2} = -2.5$, so x drifts away towards $-\infty$ and a return to 0 is far from certain.

The family can only be back at 0 after $7m$ births, m of them boys, which happens with probability $\binom{7m}{m}/2^{7m}$. So the expected number of times x is at 0, counting the start, is

$$V = \sum_{m=0}^{\infty} \frac{\binom{7m}{m}}{2^{7m}} = 1 + \frac{7}{128} + \frac{91}{16384} + \ldots \approx 1.06096$$

Now let p be the chance of ever coming back to 0. Every time x is at 0 the future looks the same as at the start, so x visits 0 exactly k times with probability $p^{k-1}(1-p)$, and the expected number of visits is $V = \frac{1}{1-p}$. Therefore

$$p = 1 - \frac{1}{V} \approx 0.05746$$

Most of it is the very first chance: 6 girls and 1 boy in the first 7 births has probability $\frac{7}{128} \approx 5.47\%$, and all the later chances together add only about 0.28%. (The sum has a closed form: $p = \frac{7(2r-1)}{2r}$, where $r \approx 0.50414$ is the root of $r^7 - 2r + 1 = 0$ between $\frac{1}{2}$ and 1.)

8.2 Circle of dwarves

N dwarves stand in a circle and a black or a white hat is placed on the head of each, at random. No dwarf can see his own hat but every dwarf sees the hats of all the others. No communication is possible once the hats are on. At some moment all the dwarves must shout, at the same time, the color of their own hat. They want to maximize the chance that *all* of them are right. What strategy should they plan beforehand? How good can it be?

8.2.1 Solution

They agree to assume that the total number of black hats is even. Each dwarf counts the black hats he sees. If the count is even he shouts "white" and if it is odd he shouts "black", which is exactly the color that makes the total even.

If the total number of black hats is indeed even then all the dwarves are right, and if it is odd then all of them are wrong. The total is even with probability 1/2, so they all survive with probability 1/2.

This cannot be improved: the first dwarf alone is right with probability 1/2, since he knows nothing about his own hat, so all of them together cannot be right more often than that.

It seems strange that N dwarves, each of whom guesses right only half of the time, are all right half of the time. The trick is that the strategy does not make anyone guess better. It makes the mistakes happen together: whenever one of the dwarves is wrong, all of them are wrong.

8.3 A letter to a hundred people

You get a letter that says: "This letter was sent to exactly 100 people and you are one of them. You may reply to the return address or not. If I get exactly one reply, each of the 100 people will get a million dollars. If I get no reply, or more than one reply, nobody gets anything."

You cannot communicate with the other 99 people, and they are all at least as smart as you are. What should you do to maximize the chance of winning?

8.3.1 Solution

There is no way to break the symmetry between the 100 people: everybody is in exactly the same situation and is as smart as you, so whatever you decide, they decide too. If you decide to reply then everybody replies, and if you decide not to reply then nobody does. Either way nobody wins.

The way out is to use randomness: reply with some probability p, for example by rolling dice. Everybody else will compute the same p. The letter writer gets exactly one reply if one person replies and the other 99 do not. There are 100 ways to choose that person, so the chance of winning is

$$f(p) = 100p(1-p)^{99}$$

To find the maximum, take the derivative:

$$f'(p) = 100(1-p)^{99} - 100 \cdot 99p(1-p)^{98} = 100(1-p)^{98}(1-p-99p)$$

It is zero when $1 - 100p = 0$, that is, at $p = \frac{1}{100}$. So intuition is right: each person should reply with probability $\frac{1}{100}$, so that on average exactly one person replies.

The chance of winning is then $\left(1 - \frac{1}{100}\right)^{99} \approx 0.370$, which is surprisingly high. With n people the chance is $\left(1 - \frac{1}{n}\right)^{n-1}$, which tends to $\frac{1}{e} \approx 0.368$ as n grows. So even with a million people the chance of winning is still almost 37%.

8.4 Monty Hall Problem

Behind 1 of 3 closed doors is a prize. You pick one of the doors. Monty opens one of the other doors and it is empty. You are given the choice of sticking with

your choice or switching to the other unopened door. Should you switch? What are your chances of winning if you do?

8.4.1 Solution

You should switch. Your chances are 2/3 if you do. The initial chances of your first choice were 1/3, and opening another door without the prize doesn't actually change that, so the remaining door now has a 2/3 chance because the chances of all possibilities must sum to 1.

Many people think the chances are 50/50 because there are two doors left, but that is not correct. If you're not convinced 2/3 is correct, consider a 100 door version: you pick 1 door and Monty opens 98 other doors avoiding the one with the prize. Now you have a 99% chance if you switch.

8.5 4 Door Monty Hall Problem

Behind 1 of 4 closed doors is a prize. You pick door 1. Monty opens the 4th door and it is empty. You switch your choice to door 2. Now Monty opens door 3 and it is also empty. Given the choice, should you switch back to door 1, and what are your chances of winning if you do?

8.5.1 Solution

Yes, your chances are 5/8 if you switch. The initial chances of your first choice were 1/4, and opening another door without the prize doesn't change that, so the remaining 2 doors now have a 3/8 chance because the chances of all possibilities must sum to 1. But then when Monty opens door 3, using the same reasoning, door 2 stays at 3/8 chance and so the remaining door changes to 5/8.

8.6 Random walk

A person stands at place 0 on a line. Every second he jumps one place to the right or one place to the left, with probability 1/2 each. What is the chance that he reaches -1 before he reaches 100?

8.6.1 Solution

The answer is $\frac{100}{101}$.

Let p_k be the chance of reaching 100 before -1 when starting at place k. Then $p_{-1} = 0$ and $p_{100} = 1$, and for every place in between the first jump goes to $k-1$ or $k+1$ with equal chances, so

$$p_k = \frac{p_{k-1}+p_{k+1}}{2}$$

This says that $p_{k+1} - p_k = p_k - p_{k-1}$: the steps between consecutive values are all equal, so p_k grows linearly from 0 at -1 to 1 at 100. There are 101 steps between -1 and 100, so $p_k = \frac{k+1}{101}$, and in particular $p_0 = \frac{1}{101}$.

The walk reaches one of the two ends with probability 1, so the chance of reaching -1 first is $1 - \frac{1}{101} = \frac{100}{101}$.

Another way to see it: the game is fair, so on average the person's position never changes. If he reaches 100 with chance q and -1 with chance $1 - q$ then his average final position is $100q - (1 - q) = 0$, so $q = \frac{1}{101}$.

8.7 Three Cards

There are 3 cards: one is all red, one is all blue, and the third is blue on one side and red on the other. The cards are shuffled. You pick one at random and it is blue on the side facing you. What are the chances that it is also blue on the other side?

8.7.1 Solution

2/3. 2 of the 3 blue sides have blue on the other side. You might think the odds are 50/50 because you could have one of two cards, but you actually get a side at random, not a card, so you are more likely to have the blue/blue card than the red/blue card.

Here is another way to think of it: When you pick a card at random, the chances of it being the same on both sides (either all red or all blue) are 2 out of 3, and this doesn't change just because you see one of the sides. If they did, they'd change no matter which color you saw, and it wouldn't even matter if you looked. So when you do see the blue side, the red/red card is eliminated, but the blue/blue card still has a 2/3 probability.

8.8 Two coin tossers

Two players play a game made of rounds. In every round each of them tosses a fair coin and sees only the result of his own coin, and then each of them guesses the result of his partner's coin, without any communication. A round is won if at least one of them guesses right, and lost if both are wrong. They can agree on a strategy before the game starts. What strategy wins the most rounds?

8.8.1 Solution

At first it seems that they cannot do better than 75%: each guess is right with probability 1/2, so if they guess independently they are both wrong with probability 1/4. But they can win every single round.

One player guesses that his partner's coin is the same as his own coin, and the other player guesses that his partner's coin is the opposite of his own coin.

In every round the two coins are either the same or different. If they are the same, the first player is right, and if they are different, the second player is right. So in every round exactly one of them is right.

Each player is still right only half of the time, as he must be, since his partner's coin is independent of everything he knows. The strategy simply arranges that their mistakes never happen in the same round.

8.9 Two Envelopes, Also known as the exchange paradox

There are two envelopes, one contains twice as much money as the other. You pick one at random and find x dollars inside. What is the "expected value" of the amount in the other envelope? Given the option, should you switch? Does it matter how much you found in that first envelope? Does it matter if you open the envelope?

8.9.1 Solution

This is a strange paradox because it appears that the expected value is the average of $2x$ and $1/2x$ which are equally probable. The result is therefore $1.25x$ so you should switch. However, this would be true for any value of x you find, so it wouldn't even matter if you open the envelope or not. This same logic suggests that you should also immediately switch back to the original envelope again, which doesn't make sense at all.

The fault is in the wrong working out of the expectation. The x in each option is not the same x since the sums $1/2x$, x and $2x$ do not occur simultaneously. Let's do the calculation more carefully. We will denote the envelopes with the sums $1/2x$ and x. With probability of $1/2$ you got the envelope with $1/2x$ which means you $1/2x$ to gain if you switch. With the same probability of $1/2$ you got the envelope with x in which case you have $1/2x$ to lose if you switch. This means that the expected value of switching is 0 as it should be. No paradox at all. If we had denoted the two envelopes x and $2x$ we would have got the same result: with probability $1/2$ we got the envelope with x in it and can gain x if we switch. With the same probability we got the envelope with $2x$ and stand to lose x if we switch. The overall expected value is again 0 as it should be.

Chapter 9

Programming

9.1 Array with constant time set all

Implement an array of a known size n with the following API:

```
set(i, v)   - set the i'th element to v
get(i)      - return the value of the i'th element
set_all(v)  - set all the elements to v
```

All three operations must run in $O(1)$ time.

9.1.1 Solution

Store a version number with every element, in addition to its value, and keep a global version and a global value. An element whose version is the global version holds its own value. Any other element was overwritten by the last `set_all` and its value is the global value.

```
init():
    global_version = 1
    global_value = 0
    allocate value[n] and version[n], all versions 0

set(i, v):
    value[i] = v
    version[i] = global_version

get(i):
    if version[i] == global_version:
        return value[i]
    return global_value
```

```
set_all(v):
    global_version += 1
    global_value = v
```

`set_all` does not touch the elements at all. It just makes every element stale by moving the global version forward.

There is one problem with this solution: the version is stored in a finite integer which will eventually wrap around. After 2^{32} calls to `set_all` (for a 32 bit version) the global version is back to an old value and an element that was set long ago suddenly looks fresh again.

To solve this, every operation also does a little cleanup work. We keep a cursor that walks over the array cyclically and every operation advances it by one element. If the element under the cursor is stale (its version is not the global version) we set its version to `global_version-1`. That element stays stale, so this does not change the result of any `get`. But now every element is visited at least once every n operations, so a stale version is never more than n versions behind the global version. As long as n is smaller than 2^{32} the global version can never wrap around to the version of a stale element. All operations are still $O(1)$.

9.2 Barrier using semaphores

Implement a barrier for n threads using semaphores. A thread that calls `barrier()` must sleep until all n threads have called it, and then all of them continue. The barrier must be reusable: the same threads will call it again and again.

9.2.1 Solution

We use three semaphores: `mutex` protects the counter, `sleep` is where the threads wait and `done` lets the last thread know that everybody woke up. `up(s, k)` and `down(s, k)` add and subtract k from a semaphore (as `semop(2)` does).

```
init(n):
    set(mutex, 1)
    set(sleep, 0)
    set(done, 0)
    count = 0

barrier():
    down(mutex, 1)
    count += 1
    if count == n:
        # the last thread to arrive wakes everybody up
        count = 0
```

```
        up(sleep, n-1)
        down(done, n-1)
        up(mutex, 1)
    else:
        # order matters: release the mutex and only then sleep
        up(mutex, 1)
        down(sleep, 1)
        up(done, 1)
```

A sleeping thread must release the mutex before it goes to sleep, otherwise no other thread could ever reach the barrier.

Why does the last thread wait on `done` while holding the mutex? Without it, a fast thread that was just woken could run ahead, call `barrier()` again for the next round and grab one of the $n-1$ wake up tokens of the current round. It would then pass the barrier without waiting for the next round, and one of the threads of the current round would stay asleep. Holding the mutex until all $n-1$ threads have taken their tokens means that no thread can start the next round before the current round is over.

9.3 LRU cache

Design a cache that holds at most n elements and, when it is full, throws away the least recently used element. Looking up an element and adding an element should both take $O(1)$ time.

9.3.1 Solution

Use a hash table together with a doubly linked list. The list holds the elements in the order in which they were used: the least recently used at the head and the most recently used at the tail. The hash table maps each key to its node in the list, so we can find any element in $O(1)$.

```
access(key):
    if key in hash:
        node = hash[key]
        cut node out of the list
    else:
        if size of the list == n:
            remove the node at the head of the list
            and remove its key from the hash
        node = new node for key
        hash[key] = node
    add node at the tail of the list
```

The list must be doubly linked so that a node can be cut out of the middle of the list in $O(1)$, given only a pointer to it.

9.4 Serving a map from many servers

A farm of servers supplies a map application with data about streets and so forth. The whole map does not fit on a single server, so the map has to be distributed among the servers. Every request of the application asks for the data of a square of 1 mile by 1 mile, anywhere on the map. To keep the application fast, every request must be answered by a single server. How would you distribute the data among the servers?

9.4.1 Solution

If we just cut the map into disjoint squares, one per server, a request that falls on the border between two squares (or on the corner of four squares) would need several servers. So the regions of the servers must overlap.

Let each server hold a square of side s miles, and place these squares on a grid with a step of $d = s - 1$ miles: the server (i, j) holds the square $[id, id + s] \times [jd, jd + s]$. Neighboring squares overlap by exactly 1 mile.

A request for the square whose lower left corner is (x, y) is sent to server $(\lfloor x/d \rfloor, \lfloor y/d \rfloor)$. Indeed, if $i = \lfloor x/d \rfloor$ then $id \le x < id+d$, so $x+1 < id+d+1 = id+s$ and the whole range $[x, x+1]$ lies inside the range $[id, id+s]$ of the server. The same goes for y. Finding the server is a simple computation and needs no lookup at all.

The price is that the data is duplicated: every server holds s^2 square miles but is responsible for only d^2 of them, so the total storage is multiplied by $\left(\frac{s}{s-1}\right)^2$. The bigger each server's square, the smaller the waste: with $s = 2$ the data is stored 4 times, with $s = 11$ only 1.21 times. So each server should hold as big a square as it can.

In practice the map is not uniform: a square mile in a city holds much more data and gets many more requests than a square mile of desert. So instead of a uniform grid, the map can be cut into regions of different sizes (for example with a quad tree), small ones in dense areas and big ones in sparse areas, as long as each server also holds the 1 mile strip to the right of and above its region, and every request goes to the region that holds its lower left corner. Popular regions can also be replicated on several servers to share the load.

9.5 Seven and nine without if

Write a function that accepts a number, which is either 7 or 9. If the number is 7 it returns 9, and if the number is 9 it returns 7. Do not use an `if` statement (or any other kind of condition).

9.5.1 Solution

There are several ways. Arithmetic: `return 16 - x`, since $7 + 9 = 16$.

Logic: `return x ^ 14`, where `^` is the bitwise exclusive or. Since $7 \oplus 9 = 14$, we get $7 \oplus 14 = 9$ and $9 \oplus 14 = 7$. In binary, $7 = 0111$, $9 = 1001$ and $14 = 1110$.

A data structure: `return {7: 9, 9: 7}[x]`, a lookup table.

The first two use the same idea: for any two numbers a and b, both $a + b - x$ and $a \oplus b \oplus x$ swap a and b.